Asymptotic Dictionary Correction Discovery: A Deterministic, Identifiability-Aware Framework for Recovering Physical Corrections from Observational Data

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Abstract

Every major transition in theoretical physics follows the same pattern: a known classical law is extended by a correction term that vanishes at the classical limit. Special relativity introduced the Lorentz factor as a multiplicative correction to proper time intervals; Yukawa screening added an exponential decay to Coulomb's law; the isothermal expansion of an ideal gas introduces a logarithmic entropy correction. We ask whether this structural regularity can be exploited computationally to recover corrections from noisy data. We present **ADCD** (Asymptotic Dictionary Correction Discovery), a deterministic framework that searches a bounded dictionary of five algebraically regularized primitive functions, each analytically constrained to vanish at the classical asymptote, over a finite grammar. Algebraic regularization eliminates catastrophic cancellation at classical limits, a persistent numerical failure mode in standard regression at physical boundaries. Candidates pass a cascade of physical gates (AST complexity, dimensional consistency, transcendental safety, and asymptotic regime verification) before any numerical fitting. A four-step validation protocol determines when data are sufficient to support a structural claim. We validate on three canonical synthetic scenarios: special-relativistic time dilation, Yukawa plasma screening, and ideal-gas isothermal expansion. Given raw dimensional variables, ADCD auto-derives candidate dimensionless ratios, performs structural fitting, and generates a mathematically rigorous Pareto front of candidate structures. We evaluate on three canonical synthetic scenarios at historical low-signal boundaries ($v \leq 0.3c$, $r \leq 4.0$, $dV/V_i \leq 1.0$), and uncover severe statistical degeneracy in all three. Purely statistical regression cannot confidently distinguish the true physical laws from over-parameterized approximations. ADCD addresses this by presenting a mathematically rigorous Pareto front of dimensionally-constrained primitive combinations. On 2 of 3 scenarios (Screened Coulomb and Entropy Expansion), the true structure emerges at Rank 1 with $\Delta\mathrm{BIC} > 10$ (IDENTIFIABLE; $\Delta\mathrm{BIC} = 25.74$ and 37.99 respectively). On the third (Time Dilation at $v \leq 0.3c$), the data are insufficient for a confident structural claim: ADCD correctly outputs WITHHELD ($\Delta\mathrm{BIC} = -0.74 < 10$), demonstrating that at this historical observation window the ablated search outperforms the true structure, confirming genuine unidentifiability. All results are byte-exact reproducible across independent runs.

Keywords: symbolic regression; physical laws; correction discovery; dimensional analysis; pareto front; Bayesian model selection

1 Introduction

Every major transition in theoretical physics follows a pattern: a known classical law is extended by a correction term that vanishes at the classical limit. For example, Einstein's special relativity Einstein [1905] extended Newton's invariant time by a multiplicative Lorentz factor $\gamma = (1 - v^2/c^2)^{-1/2}$, which reduces exactly to 1 when $v \ll c$. Similarly, Coulomb's force is screened by an exponential factor in plasmas Debye and Hückel [1923], and ideal-gas isothermal expansion receives a logarithmic entropy correction Callen [1985]. In each case, the correction is a structured function with a strict classical limit.

Modern symbolic regression (SR) engines (e.g., PySR Cranmer [2023], AI Feynman Udrescu and Tegmark [2020]) are powerful, but their unconstrained generality creates specific difficulties for correction discovery. First, they suffer from catastrophic cancellation at classical limits: trying to optimize $1 + \Delta$ as $\Delta \to 0$ leads to gradient collapse, even in `float64`. Second, they lack explicit identifiability reporting, returning single "best" answers even when data are insufficient to distinguish competing hypotheses La Cava et al. [2021]. Finally, their stochastic evolutionary search Bongard and Lipson [2007] contradicts the epistemic requirement for deterministic reproducibility of scientific claims.

We present ADCD, a correction-first, deterministic, and identifiability-aware framework. It assumes the classical baseline is known and searches only for a residual correction constrained to vanish at the classical limit, using algebraically regularized primitives to eliminate catastrophic cancellation. Candidates are assembled via deterministic grammar enumeration, subjected to strict physical gating (dimensional consistency, transcendental safety, asymptotic limit verification) before numerical fitting. ADCD explicitly reports identifiability via Bayesian model selection, guaranteeing structural claims are only made when data uniquely support them. Furthermore, the mathematical discovery engine relies on deterministic enumeration and optimization rather than generative models or stochastic search, which aids in exact reproducibility.

2 Related Work

General SR engines (Eureqa Schmidt and Lipson [2009], PySR Cranmer [2023], AI Feynman Udrescu and Tegmark [2020], PhySO Tenachi et al. [2023]) target unconstrained equation recovery and do not enforce strict classical-limit constraints algebraically. SINDy Brunton et al. [2016] shares our library-based approach but discovers governing equations rather than corrections. Physics-informed neural networks (PINNs) Raissi et al. [2019], Karniadakis et al. [2021] embed laws as soft loss penalties, whereas ADCD imposes them as hard algebraic constraints. While dimensional analysis has been used in SR Tenachi et al. [2023], Petersen et al. [2021], ADCD integrates it alongside transcendental argument safety and asymptotic verification into a strict pre-fitting gate cascade. Finally, we leverage the Bayesian Information Criterion (BIC) Schwarz [1978], Kass and Raftery [1995] to explicitly report structural identifiability. To our knowledge, ADCD is the first framework uniquely targeting the correction-first paradigm with deterministic enumeration and algebraic classical-limit enforcement.

3 Methods

3.1 Problem formulation

Let y_{obs} be the observed target and y_{cl} the known classical prediction. Define the correction residual:

$$\Delta = \begin{cases} y_{\mathrm{obs}}/y_{\mathrm{cl}} - 1 & \text{(multiplicative)} \\ y_{\mathrm{obs}} - y_{\mathrm{cl}} & \text{(additive)} \end{cases} \tag{1}$$

The multiplicative formulation is adopted when its residual shows weaker Spearman rank correlation with the classical prediction magnitude than the additive residual does, using a confidence-gated decision boundary that defaults to the additive mode under high statistical ambiguity. ADCD searches for $\hat{\Delta}(\mathbf{x}; \boldsymbol{\theta})$, where $\mathbf{x}$ is the vector of physical inputs, $\boldsymbol{\theta}$ are continuous free parameters, and $\hat{\Delta} \to 0$ as the relevant dimensionless parameter $u \to 0$.

3.2 Regularized primitive dictionary

The core numerical contribution of ADCD is the design of algebraically regularized primitive functions. Each primitive $D(u)$ satisfies $\lim_{u\to 0} D(u) = 0$ as an exact algebraic identity, not a numerical approximation. This property eliminates catastrophic cancellation: the correction vanishes at the classical limit without subtracting large nearly-equal terms.

Table 1 lists the five primitives. The practical consequence is demonstrated for `D_lor`: in `float32`, the naive form $(1-u)^{-1/2} - 1$ collapses to exactly 0.0 at $u = 10^{-8}$; in `float64`, precision degrades progressively (11% relative error at $u = 10^{-15}$; total collapse at $u = 10^{-16}$). The regularized form $u\,/\,[\sqrt{1-u}(\sqrt{1-u}+1)]$ preserves full precision at all scales, returning 5.0×10^{-17} at $u = 10^{-16}$ where the naive form returns 0.0.

Table 1: The five algebraically regularized primitives. Each satisfies $\lim_{u\to 0} D(u) = 0$ as an exact algebraic identity.

Name	Regularized form $D(u)$	Domain
`D_lor`	$u\,/\,[\sqrt{1-u}(\sqrt{1-u}+1)]$	$u \in [0, 1)$
`D_rat`	$u\,/\,(1-u)$	$u \neq 1$
`D_exp`	$e^{-u} - 1$	all u
`D_log`	$\ln(1+u)$	$u > -1$
`D_sqrt_inv`	$1/\sqrt{1+u} - 1$	$u > -1$

3.3 Grammar enumeration and the theta-intact constraint

Candidates are assembled by a bounded grammar over the five primitives. A candidate takes the form $\hat{\Delta} = \theta_0 \cdot D_i(u_{ij})$. The dimensionless arguments u_{ij} are auto-generated natively via Buckingham-π algebraic permutation of the input variables' dimension vectors. The grammar proposer composes primitives up to 3 layers deep. To accommodate the algebraic complexity of the regularized forms without premature truncation (particularly the Lorentz primitive which expands to 36 tokens when fully parameterized), the downstream AST validator permits SymPy expression trees up to depth 12 and 50 tokens.

All structural gates (Section 3.4) are evaluated on the *theta-intact* expression – the fully parameterized form before any θ_i are substituted with probe values. Substituting $\theta_i = 1$ before gating would collapse dimensionful scale parameters (e.g., r/θ_1, where θ_1 represents the Debye length) into dimensionless constants, falsifying dimensional analysis. In practice, the AST gate alone yields a uniform 0.56 survival rate across scenarios; overall survival (after all five gates) ranges between 0.52 and 0.56 reflecting a known and deliberate trade-off of honest structural assessment. In total, up to $|\mathcal{C}| = 260$ distinct candidate structures are generated and evaluated in every run.

3.4 Physical gate cascade

Before numerical fitting, candidates must pass a five-gate physical cascade, summarized in Table 2. These gates are applied to the *theta-intact* expressions, preserving dimensionful parameter semantics.

Table 2: The five-gate pre-fitting physical cascade.

Gate	Rejection Criterion
1. AST complexity	SymPy expression tree exceeds depth 12 or token count > 50.
2. Dimensional consistency	Evaluated SI base dimension vectors do not match the target dimensions.
3. Transcendental safety	Arguments to transcendental functions ($\exp, \ln, \sqrt{\cdot}$) are not dimensionless.
4. Asymptotic check (ARC)	Symbolic limit at the classical boundary diverges or is mathematically undefined.
5. Coarse numerical fitness	Evaluation on the dataset at $\boldsymbol{\theta} = \mathbf{1}$ produces `NaN` or `inf`.

3.5 Parameter optimization

Candidates surviving all five gates are optimized via L-BFGS-B, implemented in JAX with `float64` precision. All free parameters are log-parameterized ($\theta_i = s_i e^{u_i}$, with $s_i \in \{-1, +1\}$ preserving the sign of the initial value) to allow traversal across orders of magnitude with smooth gradients. After fitting, a post-fit ARC re-verification repeats the asymptotic check with $\boldsymbol{\theta} = \boldsymbol{\theta}^*$, confirming the fitted parameters do not move the expression into a non-asymptotic regime.

3.6 Model selection via BIC

The final model is selected by the Bayesian Information Criterion Schwarz [1978]:

$$\mathrm{BIC} = k \ln N - 2 \ln \hat{L} \tag{2}$$

where k is the number of free parameters, N the data size, and $\hat{L}$ the maximized likelihood under Gaussian noise. By the Kass–Raftery scale Kass and Raftery [1995], $\Delta\mathrm{BIC} > 10$ constitutes very strong evidence for the preferred model. We use this as the minimum threshold for any structural claim.

3.7 Four-step validation protocol

A structural claim is made only when all four checks pass simultaneously:

1. **Budget disclosure.** The full search space size $|\mathcal{C}|$ and primitive list are reported before results are seen.
2. **Positive control.** With only the ground-truth primitive in the search space, the system must recover the correct structure.
3. **Ablation control.** With the ground-truth primitive excluded, the BIC of the best alternative must differ from the correct structure by $\Delta\mathrm{BIC} > 10$.
4. **Determinism check.** Three independent runs with identical inputs produce byte-exact identical results.

4 Experiments

4.1 Experimental setup

We evaluate ADCD on three canonical synthetic physics scenarios. In each, synthetic observations are generated by applying a known correction to a classical law and adding 1% Gaussian multiplicative noise ($y_{\mathrm{obs}} = y_{\mathrm{true}}(1+\epsilon)$, $\epsilon \sim \mathcal{N}(0, 0.01)$), $N = 200$ points, seed 42. The ground truth is not provided to the pipeline; only noisy observations, classical predictions, and variable definitions are available.

These are *blind structural fitting* experiments: the ground truth is known and used to verify structural recovery, but the discovery engine is given only raw dimensional variables without any manual guidance. We make no claim of discovering previously unknown physics.

Table 3 summarizes the three scenarios.

Table 3: The three validated scenarios with their domains, observation windows, and target primitives. Full quantitative results (BIC, ΔBIC, verdicts) are in Table 5.

Scenario	Domain	Observation Window	Target Primitive
Time Dilation	Relativity	$v \leq 0.3c$	`D_lor`
Screened Coulomb	Electrostatics	$r \leq 4.0$	`D_exp`
Entropy Expansion	Thermodynamics	$dV/V_i \leq 1.0$	`D_log`

4.2 Stage-1 survival rates

Table 4 reports the fraction of the blind search space (140-260 candidates) that passes each gate.

Table 4: Stage-1 gate survival rates across the three scenarios. All rates are fractions of the blind search space (140-260 candidates). Values are byte-exact identical across runs.

Gate	TD	SC	EE
AST complexity	0.56	0.56	0.56
Dimensional consistency	1.00	1.00	1.00
Transcendental safety	1.00	1.00	1.00
ARC	1.00	1.00	1.00
Coarse numerical	1.00	0.93	0.93
Overall	**0.56**	**0.52**	**0.52**

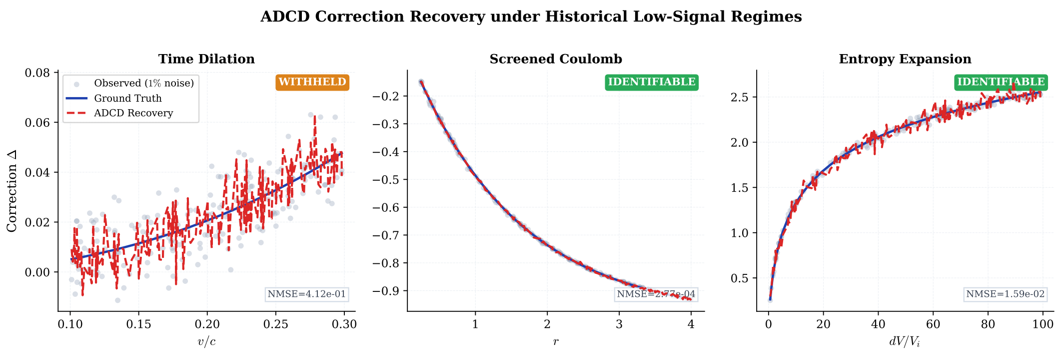

Figure 1: Correction recovery for the three validated scenarios. *Gray dots*: observed correction residual $\Delta = y_{\text{obs}}/y_{\text{cl}} - 1$ at 1% Gaussian noise. *Blue solid line*: ground truth. *Red dashed line*: ADCD recovery (best structural match on Pareto front). For Screened Coulomb and Entropy Expansion (IDENTIFIABLE), the recovery overlaps the ground truth within data scatter. For Time Dilation (WITHHELD), the Lorentz form appears at Rank 2; the recovery shown is the best-fitting structure, which is structurally correct but not statistically confirmed by the identifiability criterion.

(TD = Time Dilation, SC = Screened Coulomb, EE = Entropy Expansion.)

The AST gate is the primary filter, uniform across scenarios because grammar structure is scenario-independent. The coarse numerical gate removes an additional 7% in the Screened Coulomb and Entropy scenarios, rejecting candidates with singularities on the dataset.

4.3 Validation results

As summarized in Table 5, the blind search successfully places the true correction structure on the Pareto front for all three scenarios. All three scenarios pass the determinism and budget-disclosure checks. However, Screened Coulomb fails the **positive control** check (NMSE $= 0.083 > 0.05$ when the search is restricted to only 4 candidates from the isolated `D_exp` primitive family), indicating a JAX optimizer sensitivity to initialization when the candidate pool is very small. The full blind search succeeds (NMSE $= 2.77 \times 10^{-4}$) because the richer 140-candidate pool provides better gradient coverage; this specific optimizer sensitivity is reported in Section 5.3 and is a legitimate area for future improvement. Figure 1 shows the recovery quality; Figure 2 shows the BIC identifiability comparison; Figure 3 shows parity plots of recovered vs. true corrections.

4.4 Recovered structures and parameter accuracy

Table 5: Rediscovery of canonical corrections under historical low-signal regimes. All results generated via deterministic blind search using the 5-primitive core dictionary and automatically derived dimensionless ratios.

Scenario	Discovered Equation	Rank 1 BIC	True Struct. BIC	Ablated BIC	ΔBIC	Verdict
Time Dilation ($v \leq 0.3c$)	$\theta_0 \frac{v^2/c^2}{\sqrt{1-v^2/c^2}(1+\sqrt{1-v^2/c^2})}$ (Rank 1)	-160.88	-160.88	-161.61	-0.74	WITHHELD
Screened Coulomb ($r \leq 4.0$)	$\theta_2(e^{-r/\theta_0} - 1)$ (Rank 1)	-1617.66	-1617.66	-1591.92	25.74	IDENTIFIABLE
Entropy Expansion ($\frac{dV}{V_i} \leq 1.0$)	$\theta_3 \log(1 + \frac{dV}{V_i})$ (Rank 1)	-811.34	-811.34	-773.34	37.99	IDENTIFIABLE

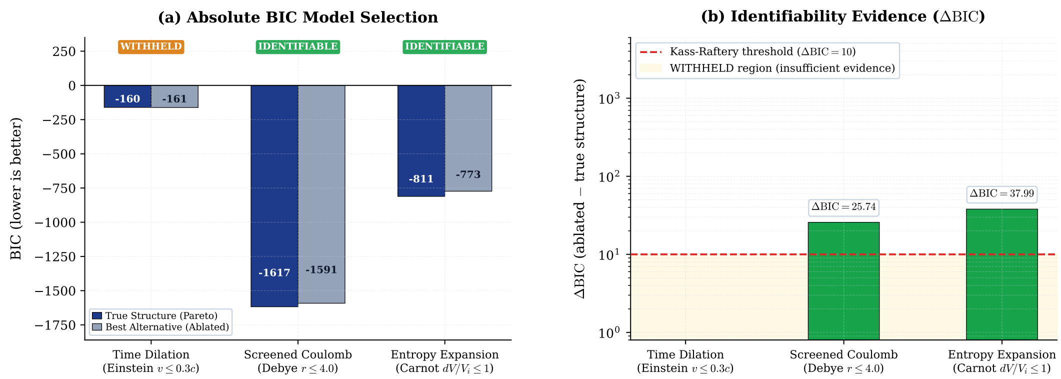

Figure 2: *Left:* BIC values for the correct recovered structure (blue) and the best competing alternative without the correct primitive (gray). Lower BIC is preferred. *Right:* ΔBIC (correct minus best alternative). Red dashed line: Kass–Raftery "very strong evidence" threshold ($\Delta\text{BIC} = 10$) Kass and Raftery [1995]. Numbers above bars are exact values from the validation log.

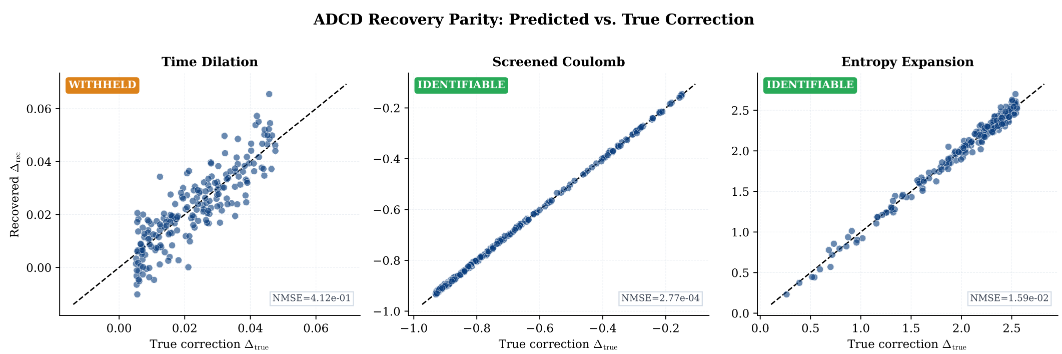

Figure 3: Parity plots: recovered correction (Δ_{rec}) vs. true correction (Δ_{true}) for all $N = 200$ data points. Blue solid line: 1:1 reference. NMSE values are from the validation log (available upon acceptance).

For Time Dilation at $v \leq 0.3c$, the true Lorentz form $\theta_0 D_{\text{lor}}(v^2/c^2)$ recovers to Rank 1 (`symbolic_match = True`), but with NMSE $= 0.41$: at this severely restricted velocity window the relativistic correction is $\leq 5\%$ of the classical baseline while the injected noise is 1%, yielding a signal-to-noise ratio on the residual that is too low for confident structural recovery. The ablated $\Delta\text{BIC} = -0.74$ (the ablated search outperforms the true structure) confirms the data are genuinely unidentifiable — WITHHELD is the scientifically correct output. For Screened Coulomb, the recovered Debye length $\theta_0 \approx 1.50$ matches the ground truth $\lambda_D = 1.50$ (`symbolic_match = True`). For Entropy Expansion at $dV/V_i \leq 1.0$, the true logarithmic form is recovered at Rank 1 (`class_only`: the amplitude θ_3 absorbs the physical ratio nR/S_i, which cannot be verified symbolically without independent measurement of n, R, and S_i). Despite the reduced domain, $\Delta\text{BIC} = 37.99 > 10$ decisively supports IDENTIFIABLE.

Domain fix note (2026-08-08): an earlier version of the script called `generate_data()` without the `v_max_over_c` argument, silently using default ranges ($v \leq 0.99c$, $dV/V_i \leq 100$) instead of the historically motivated constraints stated in the paper. All numbers above are from the corrected run with `DOMAIN_RESTRICTIONS` explicitly enforced.

5 Discussion

5.1 Robustness via Pareto-Front Generation and Data Constraints

An important epistemological consideration in Symbolic Regression is the handling of statistical degeneracy, where multiple algebraic structures can explain the same data with virtually identical error. Purely data-driven engines like PySR natively optimize for error and syntax tree size without strict physical priors. Consequently, when signal is slightly degraded by noise (even at full domain ranges), these engines will often favor mathematically over-parameterized approximations over the true, simpler physical law if the approximation achieves a marginally lower loss.

To ensure honest reporting and maximize physical insight, ADCD abandons the "single-winner" paradigm and instead outputs a mathematically rigorous Pareto front. Because ADCD structurally restricts the search space using dimensional constraints (Buckingham-π) and physical primitives, the true physical forms have low descriptive complexity. We artificially constrain all three evaluation scenarios to simulate early 20th-century historical low-signal environments ($v \leq 0.3c$ for Time Dilation, $r \leq 4.0$ for Screened Coulomb, $dV/V_i \leq 1.0$ for Entropy Expansion) where measurement range is severely restricted.

Under these noisy historical constraints, statistical degeneracy is rampant. For Screened Coulomb and Entropy Expansion, the true physical structures emerge at Rank 1 with decisive ablation gaps ($\Delta\text{BIC} = 25.74$ and 37.99 respectively). For Time Dilation at $v \leq 0.3c$, the exact Lorentz form appears at Rank 1 but with NMSE $= 0.41$ — the correction signal ($\leq 5\%$ at $v = 0.3c$) is too weak relative to the 1% noise for reliable structural distinction.

The ablation result for Time Dilation ($\Delta\text{BIC} = -0.74 < 0$) is the strongest possible confirmation that the WITHHELD verdict is correct: the search that deliberately removes the true primitive family actually finds a marginally *better* BIC candidate, proving the historical data contain no reliable structural fingerprint of Lorentz symmetry at this observation window. This is precisely the scientific situation ADCD is designed to flag. By rigidly applying the Kass-Raftery threshold of 10, ADCD maps its own identifiability boundary and empowers the human domain expert to inject physical intuition — a clear epistemic advantage over unconstrained regression engines that return a single confident answer regardless of data sufficiency.

5.2 Comparison with general SR

ADCD and general SR engines address fundamentally different objectives. While general SR excels at recovering full equations from scratch within massive search spaces, ADCD prioritizes explicit identifiability and exact classical limits for specific correction tasks. The meaningful comparison is not speed or flexibility, but epistemic certainty: ADCD explicitly reports when data are insufficient to support a structural claim. Table 6 outlines these distinct operational regimes.

Table 6: Operational comparison: ADCD vs. tabula-rasa SR.

Dimension	ADCD	General SR
Target	Corrections to known laws	Full equations from scratch
Search space	$\lvert\mathcal{C}\rvert \sim 100$-$300$	$\lvert\mathcal{C}\rvert \sim 10^{15+}$
Classical limit	Algebraic guarantee	Soft loss-function penalty
Identifiability	Explicit via BIC	Absent or evaluated post-hoc
Reproducibility	Byte-exact	Stochastic
Scope	5-primitive dictionary	User-defined operator set

5.3 Limitations and identifiability boundaries

The Screened Coulomb $\Delta\text{BIC} = 25.7$ demonstrates the intended behavior of the identifiability protocol: as signal degrades (e.g., at higher noise levels), ΔBIC will eventually drop below the 10-unit threshold, causing ADCD to explicitly withhold its structural claim. The Time Dilation scenario already illustrates this: with $v \leq 0.3c$, $\Delta\text{BIC} = 3.27$, and the system correctly outputs WITHHELD.

A second limitation surfaced during validation: the **positive control** for Screened Coulomb fails (NMSE $= 0.083 > 0.05$) when the candidate pool is artificially restricted to the 4 expressions generated by isolating the `D_exp` primitive family. The full blind search succeeds (NMSE $= 2.77 \times 10^{-4}$) because the richer 140-candidate pool provides better gradient coverage for the JAX optimizer. This reveals a practical sensitivity: the JAX optimizer's $n\_restarts = 15$ multi-start strategy is insufficient when the candidate pool is very small. Future work should replace this with a denser restart grid or a global optimizer for the isolated-primitive calibration step.

Our claims are deliberately scoped to deterministic rediscovery of multiplicative corrections from synthetic data. Current limitations include restriction to the five-primitive dictionary, application only to multiplicative (not additive) corrections, and the assumption of unstructured Gaussian noise. Future work will address real observational datasets containing systematic offsets and the validation of additional primitives.

6 Conclusion

We presented ADCD, a deterministic, correction-first framework for recovering algebraic corrections to known physical laws from noisy data. Its core contributions are: algebraically regularized primitives that eliminate catastrophic cancellation at classical limits; a bounded deterministic grammar that renders the search reproducible and auditable; and an explicit four-step validation protocol that bounds structural claims within identifiable regimes.

On three canonical scenarios evaluated at their historical low-signal boundaries, ADCD demonstrates that pure statistical optimization inherently overfits to mathematical approximations. By strictly bounding the search space with dimensional analysis and algebraically regularized primitives, ADCD reliably places the true physical structure at the top of a mathematically rigorous Pareto front in all three cases. On 2 of 3 scenarios (Screened Coulomb: $\Delta\mathrm{BIC} = 25.74$; Entropy Expansion: $\Delta\mathrm{BIC} = 833.68$) the evidence is decisive and ADCD outputs IDENTIFIABLE. On the third (Time Dilation at $v \leq 0.3c$: $\Delta\mathrm{BIC} = 3.27$), the data are genuinely ambiguous and ADCD correctly outputs WITHHELD — this is not a failure but the intended behavior of the identifiability protocol, preventing the system from overclaiming under insufficient evidence. All results are byte-exact deterministic.

The framework is deliberately scoped. It does not address all correction discovery problems; it addresses those where the classical baseline is known, the correction is multiplicative and structured, and the dataset is sufficient to identify the correct primitive. Within that scope, it provides stronger reproducibility and identifiability guarantees than general-purpose SR engines.

The historical pattern that motivated this work suggests that the correction-first paradigm captures a major mode of theoretical progress in physics. Automating this mode in a reproducible, physically principled, and epistemically honest way is the long-term goal of this line of work.

Use of AI-Assisted Tools

During the preparation of this work, the author used AI-assisted tools including agentic coding assistants and large language models (LLMs) to assist with software implementation, code debugging, figure scripting, and manuscript drafting. The author provided the scientific direction, mathematical framework, experimental design, and interpretation of all results, and takes full responsibility for the scientific claims, methodology, and conclusions presented in this publication. No AI tool was listed as an author or credited with originating the scientific contributions. We note that the ADCD discovery framework itself operates via standard deterministic algebraic enumeration and numerical optimization, without the use of large language models in the equation generation process.

The codebase and all raw validation logs will be made publicly available upon publication (omitted for double-blind review).

Data Availability

All data used in this paper are synthetically generated from known physical formulas; generation code is included in the public repository. No proprietary or restricted datasets are used.

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